DATA ENGINEERING TAKE-HOME · DEEL

Open orders, a few seconds after they change

A Spark Structured Streaming pipeline from Debezium CDC to a dimensional model, with the four logistics metrics maintained incrementally and served live.

Leonardo Mariano · November 2026

THE ASK

Logistics leaders need current and historical order state, with low latency

History + current state

Debezium snapshot and change stream land in one model: latest state in the facts, every status, delivery date or customer change in a history fact, every customer and product version in SCD2 dimensions.

Dimensional model

Date, status, customer and product dimensions; order and order-line facts; a transactional change-history fact; one mart table per question.

As soon as possible

5 s trigger. Measured source-commit to mart-commit latency in steady state: **6.9 s average, 7.5 s p95** (8.0 s p95 for the oldest event of a batch).

Q1
Open orders by delivery date and status

Q2
Top 3 delivery dates with more open orders

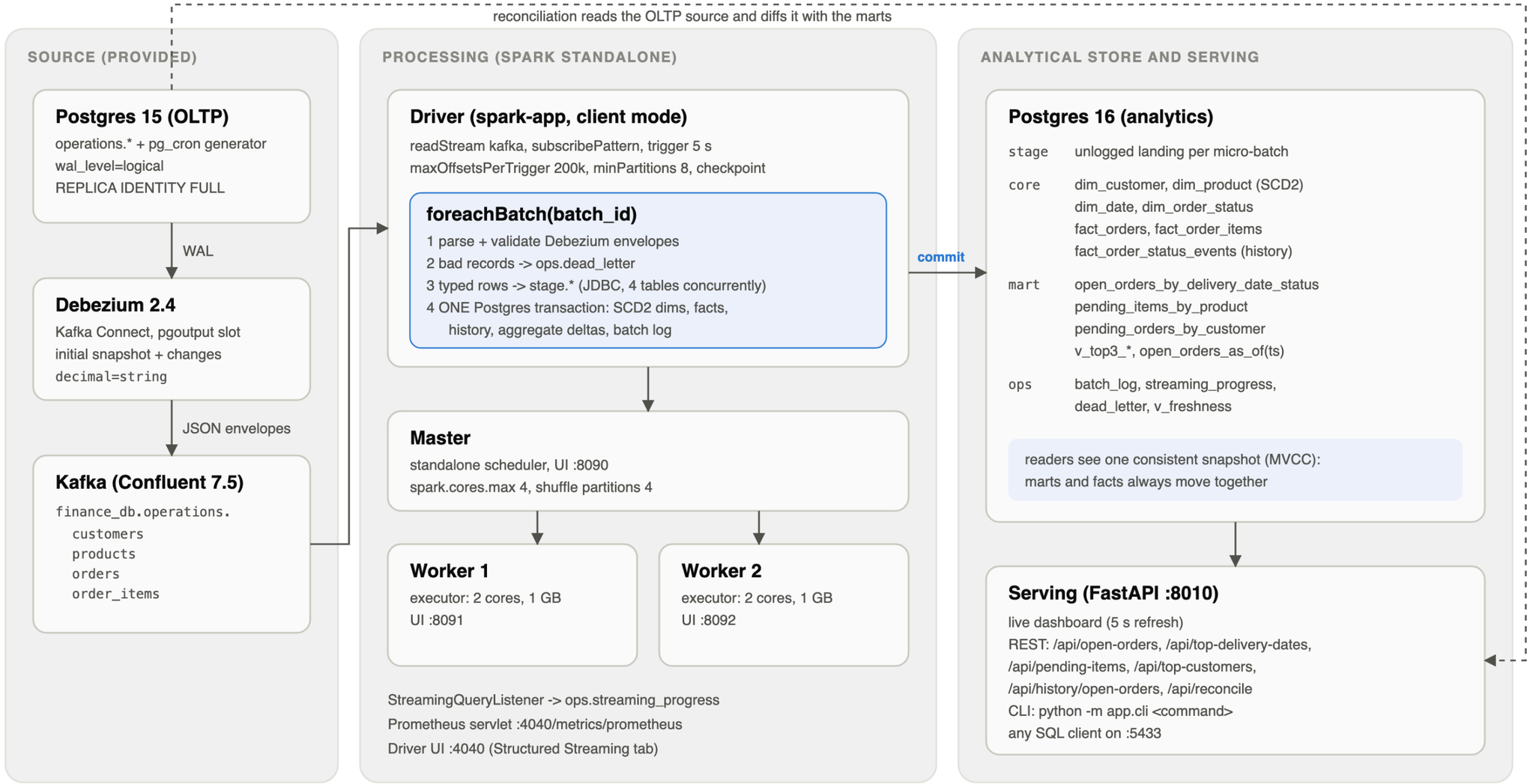
Q3
Open pending items by product

Q4
Top 3 customers with more pending orders

Open = status is not COMPLETED and not deleted. Rule stored in dim_order_status.

ARCHITECTURE

One streaming query, one transaction per micro-batch



Everything runs with docker compose up; source stack is the one provided

INSIDE FOREACHBATCH

Spark parses and stages, Postgres applies atomically

1

Parse and validate

Debezium envelope per topic. Row image is `after`, or `before` for deletes. Bad records go to `ops.dead_letter` with the raw payload.

2

Stage

Typed rows into unlogged `stage.*` tables over JDBC. The four tables are staged concurrently and Kafka ranges are split into 8 tasks. All versions kept.

3

Apply in one transaction

SCD2 dimensions, version-guarded fact upserts, status history, aggregate deltas and the batch log commit together.

Why a single query over four topics

One checkpoint and one transaction for all tables. An order and its items may still land in different batches; the merge handles either order of arrival.

Why Spark holds no state

The join with history happens against the persisted model. No state store, watermark or stream-stream join; history survives Kafka retention and checkpoint loss.

Why one transaction

Readers get a consistent MVCC snapshot: an aggregate never disagrees with the facts behind it.

spark/app/pipeline.py, transform.py, merge_sql.py

DIMENSIONAL MODEL

Current state, full history and serving tables in one schema family

Table	Grain	Role
core.dim_date	day	calendar attributes for delivery and order dates
core.dim_order_status	status	carries the business rule `is_open`
core.dim_customer, dim_product	version (SCD2)	valid_from / valid_to / is_current; keeps price and address history
core.fact_orders	order, latest state	accumulating snapshot with date keys
core.fact_order_items	order line, latest state	source column `quanity` fixed to `quantity`
core.fact_order_status_events	order change	the history; powers `open_orders_as_of(ts)` and order timelines
mart.* (3 tables + views)	delivery date x status, product, customer	Q1 to Q4, maintained by deltas; top 3 uses RANK so ties are visible
ops.*	batch, progress event, bad record	batch log with latency, Spark progress, dead letters, freshness view

CORRECTNESS

Exactly-once results on top of at-least-once delivery

```
delta(key) = contribution after the batch - contribution before the batch
```

Version guard

Fact upserts only win when `(source_lsn, kafka_offset)` is newer. Replays and late events are no-ops.

Deltas from real state

Before and after state are read inside the same transaction. A replayed batch changes nothing, so its delta is zero.

Unique history

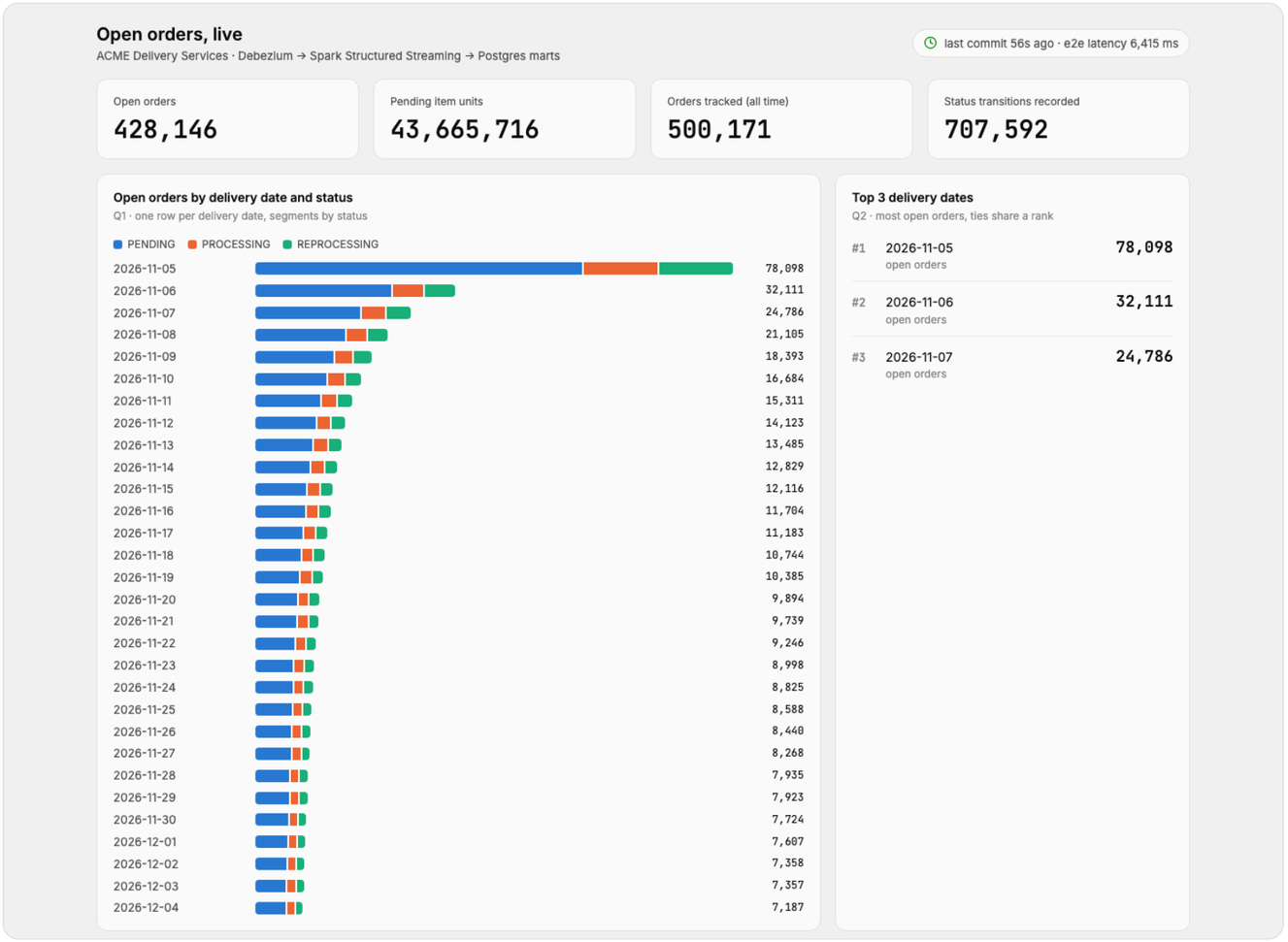
One history row per source change `(order_id, LSN)`. SCD2 only versions real attribute changes newer than the stored one.

Proven, not assumed: a 188k-event batch already committed in Postgres was forced to run again (driver killed, its checkpoint commit removed). Debezium also re-published ~719k events after a crash. In both cases the reconciliation matched the source exactly, with zero duplicate history rows. Merge tests assert the same on a throwaway database.

Items that arrive before their order contribute nothing until the order arrives, then the order change pulls them in

SERVING

Dashboard, REST API, CLI and SQL read the same marts



Live dashboard

Refreshes every 5 s, with a freshness indicator: seconds since the last commit and end-to-end latency.

REST API

```
/api/open-orders
/api/top-delivery-dates
/api/pending-items
/api/top-customers
/api/history/open-orders?as_of=
/api/reconcile
```

CLI and SQL

`python -m app.cli top-dates` and views like `mart.v_top3_customers`

FastAPI + psycopg pool, reads only mart and ops schemas

THE ANSWERS, LOAD-TEST DATA (02:14 UTC, SAME MOMENT AS THE DASHBOARD)

500k orders tracked, 428k open, 43.7M item units pending

Q2 · top delivery dates

2026-11-05	78,092
2026-11-06	32,108
2026-11-07	24,784

Q4 · top customers

Amazing Partner LLC	69,385
Very Good Company	39,563
Sky Company	36,830

Q3 · top pending products

Fridge	2,140,920
Desk	2,129,077
Pants	2,107,764

Reconciled against the source. Every key of the three aggregates (90 delivery date x status, 28 products, 18 customers) equals the same metric computed directly on the OLTP database. With the generator alone only ~20 orders stay open and they have no items (slide 11).

Volume comes from scripts/load_test.sql (300k + 100k orders) on top of the provided generator

CLUSTER SIZING

Size by source parallelism, batch volume and the sink, not by history

Variable	Observed	Decision
Source parallelism	4 topics x 1 partition	4 cores (2 executors x 2); tables staged concurrently and minPartitions 8 so no core idles; shuffle partitions 4 instead of 200
Event rate	3.6 events/s steady, bursts of 1.6M	`maxOffsetsPerTrigger = 200k` as backpressure
Event size	0.5 to 0.8 KB JSON	200k events ~150 MB cached: 1 GB executors with headroom
Latency target	seconds	5 s processing-time trigger
State	none in Spark	memory does not grow with keys or history
Sink	98% of trigger time in addBatch	merge is now the larger half; scale the sink first
Stack memory	8 GB Docker VM	JVM heaps capped after Kafka Connect was OOM-killed under load

Production rule of thumb: about 4.5k events/s per core with this sink. For a peak of R events/s plan R / 4500 × 2 cores, and at least as many Kafka partitions as cores, keyed by primary key.

Spark standalone: 1 master, 2 workers (2 cores, 1.5 GB), driver in client mode

SPARK METRICS

What the metrics said, and what they point to

processedRowsPerSecond

18.3k

avg on 200k batches (24.3k max), was 9.3k

addBatch / triggerExecution

98%

time is in foreachBatch, not in Kafka reads

stage vs merge (200k batch)

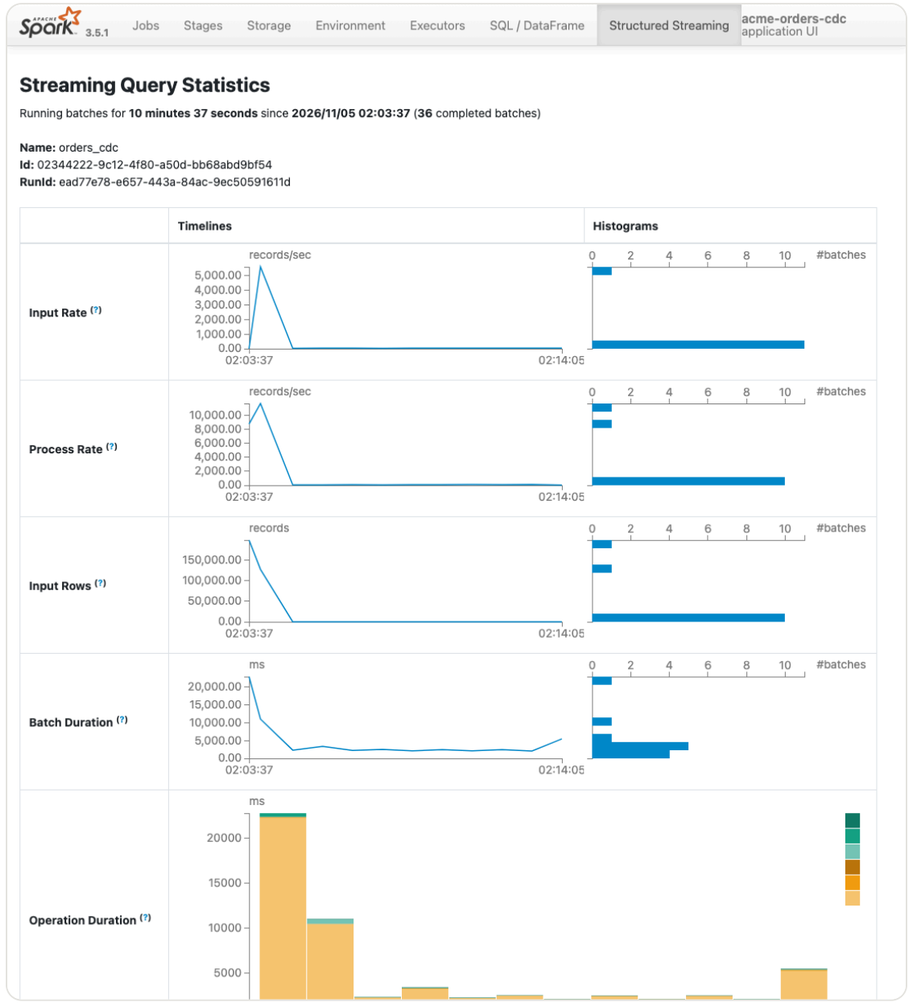
4.3 / 5.2 s

staging was 14.3 s on one core; fixed with concurrency

e2e latency, steady

6.9 s

p95 7.5 s; 1.6M burst drained in ~2 min



Collected by a `StreamingQueryListener` into `ops.streaming_progress`, by the job into `ops.batch_log`, and exposed on `:4040/metrics/prometheus`.

Clean run from a fresh clone: 3.57M Kafka records consumed, zero dead letters

THE PATH

What testing broke, and what changed

Staging used one core of four

Metrics showed 14.3 s of staging per 200k batch: four single-partition topics staged one after the other. Concurrent tables plus `minPartitions` : 4.3 s.

Kafka Connect OOM-killed under load

Default heap up to 2 GB on an 8 GB VM; CDC stopped silently. Heaps capped. On restart Debezium re-sent ~719k events; the version guards absorbed them.

The first crash test proved less than claimed

Killing the driver mid-batch only tested a rollback. Added merge tests on a real Postgres and a forced replay of a batch that was already committed.

State that did not survive a restart

Kafka had no volume and history was keyed by Kafka offset. Named volumes for Kafka and ZooKeeper; history keyed by `(order_id, LSN)` .

Each item is a commit in the repository history

FINDINGS IN THE SOURCE

Four things the data told me

Orders without items

The generator attaches items to the customer's *first* order of the day (`LIMIT 1` , no status filter). Orders created later, the ones that stay open, have no items, so with the generator alone Q3 is correct but nearly empty. Reconciliation confirms the source shows the same.

Orders that never close

Orders from previous days are never updated again, so they stay open forever. In production this is a stale-order alert.

Unconstrained NUMERIC

`unity_price` would arrive as a base64 struct. The connector now emits decimals as strings: lossless and simple to cast.

Schema drift from the ERD

`order_items.quanity` in the DDL versus `quantity` in the diagram. Mapped explicitly and fixed in the model.

Documented in the README with the queries used

NEXT STEPS

What I would do with more time

- **Faster sink:** `COPY` instead of JDBC inserts, partitioned facts so the merge touches smaller indexes
- **Contracts:** Avro + Schema Registry, compatibility checks in CI
- **Repair without restarts:** Debezium incremental snapshots via a signal table
- **Lakehouse for scale:** history and facts in Delta or Iceberg with MERGE; Postgres keeps only the small marts
- **Observability:** Prometheus + Grafana, alert when batch duration exceeds the trigger for N batches
- **Data quality:** alerts on dead letters, orders without items and reconciliation drift; freshness as an SLO
- **Operations:** Kubernetes, checkpoint on object storage, resource limits per service, driver exits when executors are lost
- **CI:** compose up, load script, assert reconciliation

Repository: README, docs/evidence, scripts/load_test.sql